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The pressure - volume graph for an isochoric process is a vertical line parallel to pressure axis as shown in Figure 8.37.

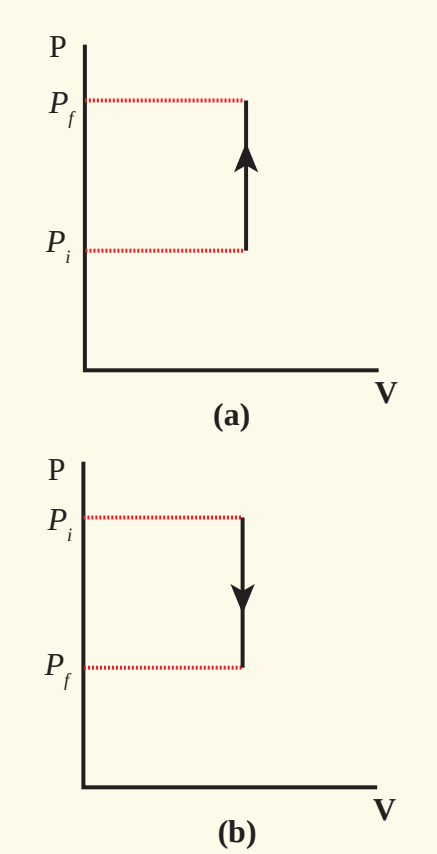


Figure 8.37 Isochoric process with (a) increased pressure and (b) decreased pressure

The equation of state for an isochoric process is given by

$$P = \left(\frac{\mu R}{V}\right) T \qquad (8.51)$$

where $\left(\frac{\mu R}{V}\right)$ = constant

We can infer that the pressure is directly proportional to temperature. This implies that the P-T graph for an isochoric process is a straight line passing through origin.

If a gas goes from state (P_i, T_i) to (P_f, T_f) at constant volume, then the system satisfies the following equation

$$\frac{P_i}{T_i} = \frac{P_f}{T_f} \qquad (8.52)$$

For an isochoric processes, $\Delta V=0$ and $W=0$. Then the first law becomes

$$\Delta U = Q \qquad (8.53)$$

Implying that the heat supplied is used to increase only the internal energy. As a result the temperature increases and pressure also increases. This is shown in Figure 8.38

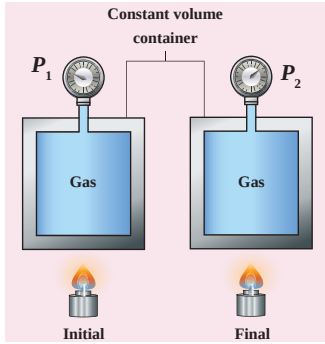


Figure 8.38 Isochoric process

Suppose a system loses heat to the surroundings through conducting walls by keeping the volume constant, then its internal energy decreases. As a result the temperature decreases; the pressure also decreases.

Examples:

1. When food is cooked by closing with a lid as shown in figure.

